DataProcessing

21 de junio de 2026

In [ ]:

```
#STEP 1:Basics of all tasks
import pandas as pd #library to see data
import numpy as np #library to use numerical functions

pd.set_option('display.expand_frame_repr', False) #to display all columns in the dataframe without truncating/rapping it in many l
df = pd.read_csv('Financials4.csv') #just to read or load, no view of the data yet
print(df.head()) #see the first 5 rows of the dataframe
```

Out [ ]:

```
   Unnamed: 0     Segment  Country     Product   Discount Band   Units Sold   Manufacturing Price   Sale Price   Gross Sales
Discounts        Sales         COGS       Profit       Date  Month Number  Month Name   Year
0           0  Government   Canada    Carretera            None    $1,618.50                $3.00           NaN   $32,370.00        $-
$32,370.00   $16,185.00   $16,185.00   1/1/2014            1     January   2014
1           1  Government  Germany    Carretera            None    $1,321.00                $3.00       $20.00    $26,420.00        $-
$26,420.00   $13,210.00   $13,210.00   1/1/2014            1     January   2014
2           2   Midmarket   France    Carretera            None    $2,178.00                $3.00       $15.00    $32,670.00        $-
$32,670.00   $21,780.00   $10,890.00   1/6/2014            6        June   2014
3           3   Midmarket  Germany    Carretera            None           NaN               $3.00       $15.00    $13,320.00        $-
$13,320.00    $8,880.00    $4,440.00   1/6/2014            6        June   2014
4           4   Midmarket   Mexico    Carretera            None    $2,470.00                $3.00   $1,000,000    $37,050.00        $-
$37,050.00   $24,700.00   $12,350.00   1/6/2014            6        June   2014
```

In [3]:

```
print(df.shape) #see the number of rows and columns in the dataframe
```

Out [3]:

```
(701, 17)
```

In [4]:

```
print(df.info()) #see the data types of each column and if there are any missing values
```

Out [4]:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 701 entries, 0 to 700
Data columns (total 17 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Unnamed: 0            701 non-null    int64
 1   Segment               701 non-null    object
 2   Country               701 non-null    object
 3    Product              701 non-null    object
 4    Discount Band        701 non-null    object
 5    Units Sold           700 non-null    object
 6    Manufacturing Price  701 non-null    object
 7    Sale Price           700 non-null    object
 8    Gross Sales          699 non-null    object
 9    Discounts            701 non-null    object
 10    Sales               701 non-null    object
 11   COGS                 701 non-null    object
 12   Profit               701 non-null    object
 13  Date                  701 non-null    object
 14  Month Number          701 non-null    int64
 15   Month Name           701 non-null    object
 16  Year                  701 non-null    int64
dtypes: int64(3), object(14)
memory usage: 93.2+ KB
None
```

In [5]:

```
print(df.columns) #see the column names in the dataframe
```

Out [5]:

```
Index(['Unnamed: 0', 'Segment', 'Country', ' Product ', ' Discount Band ',
       ' Units Sold ', ' Manufacturing Price ', ' Sale Price ',
       ' Gross Sales ', ' Discounts ', '  Sales ', ' COGS ', ' Profit ',
       'Date', 'Month Number', ' Month Name ', 'Year'],
      dtype='object')
```

In [6]:

```
print(df.isna().sum()) #see the number of missing values in each column
```

Out [6]:

```
Unnamed: 0              0
Segment                 0
Country                 0
 Product                0
 Discount Band          0
 Units Sold             1
 Manufacturing Price    0
 Sale Price             1
 Gross Sales            2
 Discounts              0
  Sales                 0
 COGS                   0
 Profit                 0
Date                    0
Month Number            0
 Month Name             0
Year                    0
dtype: int64
```

In [10]:

```
print(df)
```

Out [10]:

```
     Unnamed: 0           Segment                  Country     Product   Discount Band   Units Sold   Manufacturing Price   Sale
Price     Gross Sales    Discounts           Sales           COGS       Profit        Date  Month Number  Month Name   Year
0             0        Government                   Canada   Carretera            None    $1,618.50               $3.00
NaN     $32,370.00         $-        $32,370.00      $16,185.00   $16,185.00    1/1/2014             1     January   2014
1             1        Government                  Germany   Carretera            None    $1,321.00               $3.00
$20.00      $26,420.00         $-        $26,420.00      $13,210.00   $13,210.00    1/1/2014             1     January   2014
2             2         Midmarket                   France   Carretera            None    $2,178.00               $3.00
$15.00      $32,670.00         $-        $32,670.00      $21,780.00   $10,890.00    1/6/2014             6        June   2014
3             3         Midmarket                  Germany   Carretera            None           NaN              $3.00
$15.00      $13,320.00         $-        $13,320.00       $8,880.00    $4,440.00    1/6/2014             6        June   2014
4             4         Midmarket                   Mexico   Carretera            None    $2,470.00               $3.00
$1,000,000      $37,050.00         $-        $37,050.00      $24,700.00   $12,350.00    1/6/2014             6        June   2014
..          ...               ...                      ...          ...             ...          ...                  ...
...            ...          ...             ...             ...          ...         ...          ...          ...   ...
696         696    Small Business                   Mexico    Amarilla            High      $546.00              $260.00
$300.00    $1,63,800.00   $24,570.00    $1,39,230.00    $1,36,500.00    $2,730.00   1/10/2014            10     October   2014
697         697        Government                   Mexico     Montana            High    $1,368.00               $5.00
$7.00       $9,576.00    $1,436.40       $8,139.60       $6,840.00    $1,299.60    1/2/2014             2    February   2014
698         698        Government                   Canada       Paseo            High      $723.00              $10.00
$7.00       $5,061.00      $759.15       $4,301.85       $3,615.00      $686.85    1/4/2014             4       April   2014
699         699  Channel Partners  United States of America         VTT            High    $1,806.00              $250.00
$12.00              NaN   $3,250.80      $18,421.20       $5,418.00   $13,003.20    1/5/2014             5         May   2014
700         700  Channel Partners  United States of America         VTT            High    $1,806.00              $250.00
$12.00              NaN   $3,250.80      $18,421.20       $5,418.00   $13,003.20    1/5/2014             5         May   2014

[701 rows x 17 columns]
```

So, AFTER STEP 1, we see first coloum is unnecessary and extra, Left-most number in the printout is the row index. The first column seems extra so we remove it immediately.

In [18]:

```
# Step 2: Data CLeaning

df = df.drop(columns=['Unnamed: 0'])
```

Out [18]:

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[18], line 3
```

```
      1 # Step 2: Data Cleaning
----> 3 df = df.drop(columns=['Unnamed: 0'])

File c:\Users\ROWTECH\AppData\Local\Programs\Python\Python313\Lib\site-packages\pandas\core\frame.py:5581, in DataFrame.drop(self,
labels, axis, index, columns, level, inplace, errors)
   5433 def drop(
   5434     self,
   5435     labels: IndexLabel | None = None,
   (...)
   5442     errors: IgnoreRaise = "raise",
   5443 ) -> DataFrame | None:
   5444     """
   5445     Drop specified labels from rows or columns.
   5446
   (...)
   5579             weight  1.0     0.8
   5580     """
-> 5581     return super().drop(
   5582         labels=labels,
   5583         axis=axis,
   5584         index=index,
   5585         columns=columns,
   5586         level=level,
   5587         inplace=inplace,
   5588         errors=errors,
   5589     )

File c:\Users\ROWTECH\AppData\Local\Programs\Python\Python313\Lib\site-packages\pandas\core\generic.py:4788, in NDFrame.drop(self,
labels, axis, index, columns, level, inplace, errors)
   4786 for axis, labels in axes.items():
   4787     if labels is not None:
-> 4788         obj = obj._drop_axis(labels, axis, level=level, errors=errors)
   4790 if inplace:
   4791     self._update_inplace(obj)

File c:\Users\ROWTECH\AppData\Local\Programs\Python\Python313\Lib\site-packages\pandas\core\generic.py:4830, in
NDFrame._drop_axis(self, labels, axis, level, errors, only_slice)
   4828         new_axis = axis.drop(labels, level=level, errors=errors)
   4829     else:
-> 4830         new_axis = axis.drop(labels, errors=errors)
   4831     indexer = axis.get_indexer(new_axis)
   4833 # Case for non-unique axis
   4834 else:

File c:\Users\ROWTECH\AppData\Local\Programs\Python\Python313\Lib\site-packages\pandas\core\indexes\base.py:7070, in Index.drop(self,
labels, errors)
   7068 if mask.any():
   7069     if errors != "ignore":
-> 7070         raise KeyError(f"{labels[mask].tolist()} not found in axis")
   7071     indexer = indexer[~mask]
   7072 return self.delete(indexer)

KeyError: "['Unnamed: 0'] not found in axis"
```

This comes as already droped the colum called 'unnamed: 0'

In [19]:

```
df #to see updated dataframe after dropping the column
```

Out [19]:

	Segment	Country	Product	Discount Band	Units Sold	Manufacturing Price	Sale Price	Gross Sales	Discounts	Sales	COGS	Profit
0	Government	Canada	Carretera	None	$1,618.50	$3.00	NaN	$32,370.00	$-	$32,370.00	$16,185.00	$16,185.00
1	Government	Germany	Carretera	None	$1,321.00	$3.00	$20.00	$26,420.00	$-	$26,420.00	$13,210.00	$13,210.00
2	Midmarket	France	Carretera	None	$2,178.00	$3.00	$15.00	$32,670.00	$-	$32,670.00	$21,780.00	$10,890.00
3	Midmarket	Germany	Carretera	None	NaN	$3.00	$15.00	$13,320.00	$-	$13,320.00	$8,880.00	$4,440.00
4	Midmarket	Mexico	Carretera	None	$2,470.00	$3.00	$1,000,000	$37,050.00	$-	$37,050.00	$24,700.00	$12,350.00
...	...	...	...	...	...	...	...	...	...	...	...	...
696	Small Business	Mexico	Amarilla	High	$546.00	$260.00	$300.00	$1,63,800.00	$24,570.00	$1,39,230.00	$1,36,500.00	$2,730.00
697	Government	Mexico	Montana	High	$1,368.00	$5.00	$7.00	$9,576.00	$1,436.40	$8,139.60	$6,840.00	$1,299.60
698	Government	Canada	Paseo	High	$723.00	$10.00	$7.00	$5,061.00	$759.15	$4,301.85	$3,615.00	$686.85
699	Channel Partners	United States of America	VTT	High	$1,806.00	$250.00	$12.00	NaN	$3,250.80	$18,421.20	$5,418.00	$13,003.20

	Segment	Country	Product	Discount Band	Units Sold	Manufacturing Price	Sale Price	Gross Sales	Discounts	Sales	COGS	Profit
700	Channel Partners	United States of America	VTT	High	$1,806.00	$250.00	$12.00	NaN	$3,250.80	$18,421.20	$5,418.00	$13,003.20

701 rows × 16 columns

In [20]:

```
df.info() # summary and data types of the dataframe after dropping the column
```

Out [20]:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 701 entries, 0 to 700
Data columns (total 16 columns):
 #   Column                Non-Null Count  Dtype 
---  ------                --------------  ----- 
 0   Segment               701 non-null    object
 1   Country               701 non-null    object
 2    Product              701 non-null    object
 3    Discount Band        701 non-null    object
 4    Units Sold           700 non-null    object
 5    Manufacturing Price  701 non-null    object
 6    Sale Price           700 non-null    object
 7    Gross Sales          699 non-null    object
 8    Discounts            701 non-null    object
 9     Sales               701 non-null    object
 10   COGS                 701 non-null    object
 11   Profit               701 non-null    object
 12  Date                  701 non-null    object
 13  Month Number          701 non-null    int64 
 14   Month Name           701 non-null    object
 15  Year                  701 non-null    int64 
dtypes: int64(2), object(14)
memory usage: 87.8+ KB
```

Here, some colum have missing value, like unit sold has 1 missing value and "object" means string type while int64 are integer types. Moreover, some column names contain whitespace as we see from

In [21]:

```
df.columns # see the column names
```

Out [21]:

```
Index(['Segment', 'Country', ' Product ', ' Discount Band ', ' Units Sold ',
       ' Manufacturing Price ', ' Sale Price ', ' Gross Sales ', ' Discounts ',
       '  Sales ', ' COGS ', ' Profit ', 'Date', 'Month Number',
       ' Month Name ', 'Year'],
      dtype='object')
```

In [23]:

```
df.columns = df.columns.str.strip() #removes extra whitespace (spaces) from all column names
df.columns # see the updated column names after stripping whitespace
```

Out [23]:

```
Index(['Segment', 'Country', 'Product', 'Discount Band', 'Units Sold',
       'Manufacturing Price', 'Sale Price', 'Gross Sales', 'Discounts',
       'Sales', 'COGS', 'Profit', 'Date', 'Month Number', 'Month Name',
       'Year'],
      dtype='object')
```